Meaghan Ramlakhan

meaghanramlakhan@gmail.com | 832-938-5055 | LinkedIn | GitHub | Portfolio

Education

Rice University — Houston, TX
B.S. in Computer Science; Minors in Data Science and Statistics; GPA: 3.9/4.0, Cum Laude — *May 2026*

- Honors: Distinction in Research and Creative Work (Statistics); President's Honor Roll (Spring 2025, Fall 2025, Spring 2026).

Experience

Software Engineer — June 2026 – Present
OmniScience — *Houston, TX*

- Develop Python and Django data-ingestion pipelines and backend services for Vivo, an agentic AI platform that unifies clinical-trial data and produces traceable, real-time insights.

Data Science and Bioinformatics Intern — January 2026 – May 2026
McWilliams School of Biomedical Informatics, UTHealth Houston — *Houston, TX*

- Supported biomedical data science and bioinformatics research through Python-based data processing, validation, statistical analysis, and manuscript preparation.
- Contributed to a separate forced vital capacity (FVC) research project through data analysis and manuscript development.

CPRIT BIG-TCR Cancer and Artificial Intelligence Summer Intern — June 2025 – August 2025
McWilliams School of Biomedical Informatics, UTHealth Houston — *Houston, TX*

- Engineered and optimized a five-layer MLP in Python and TensorFlow for antibody-protein binding prediction, applying cross-validation, L2 regularization, and hyperparameter tuning to achieve 85%+ accuracy.
- Extended **BRAINCELL-AID**, a biomedical brain-annotation platform, by developing Python/MySQL pipelines for automated ingestion and processing of 5,000+ neural-cell records, integrating LLM-assisted annotation and visualization tools, and co-authoring the BRAINCELL-AID manuscript.

Computer Science Teaching Assistant — August 2023 – May 2026
Rice University — *Houston, TX*

- Supported 800+ students across four computer science courses through office hours, discussion-board responses, code reviews, and assessment grading.
- In the most recent course, mentored 10 teams across two projects in Go and TypeScript, advising on concurrency, synchronization, API design, debugging, and performance optimization.

Live Instructor Technology Intern - RCEL ELITE Tech Program — May 2024 – July 2024
Rice University — *Houston, TX*

- Selected as one of 20 instructors to develop and refine course materials in machine learning and robotics; led 40+ classroom sessions with interactive instruction, feedback, grading, and attendance management.

Computer Hardware Engineer & Mission Assurance Specialist — May 2023 – August 2023
NASA L'SPACE Program - Arizona State University — *Remote*

- Collaborated on an 11-person robotic mission concept for Ceres, designing command-and-data-handling architecture, software interfaces, and mission-risk analyses aligned with NASA review standards.

Projects

Genome Short-Read Mapper | *Python, Bioinformatics, Multiprocessing* — Aug – Dec 2025

- Built a read-mapping pipeline from scratch to parse FASTA/FASTQ files, generate k-mer seeds, identify candidate alignments, validate results against ground truth, and accelerate workloads with multiprocessing.

OwlDB: NoSQL Document Database | *Go, TypeScript, REST, SSE* — Aug – Oct 2024

- Built a concurrent document database with CRUD REST APIs, skip-list indexing, schema validation, and real-time client updates through Server-Sent Events.

Locality-Sensitive Hashing Analysis | *Python, scikit-learn* — Sep – Dec 2024

- Benchmarked signed-random-projection and k-means LSH methods on 18,000 text documents, evaluating precision, query time, and memory tradeoffs.

FDA Adverse Events Analysis | *Python, Pandas, scikit-learn* — Jan 2024

- Won the Best Underclassmen Award for analyzing the FDA CAERS dataset and modeling consumer-product safety risks.

Technical Skills

Languages: Python, Java, Go, TypeScript, JavaScript, C, SQL, R, HTML/CSS
Frameworks/Libraries: Django, React, Node.js, TensorFlow, PyTorch, Pandas, NumPy, scikit-learn, PySpark
Data/Cloud: PostgreSQL, MySQL, MongoDB, Hadoop, AWS, Google Cloud Platform (GCP), Google Cloud Storage (GCS), Docker
Developer Tools: Git, GitHub, Jira, Linux, Postman, DBeaver, VS Code, IntelliJ, PyCharm, LaTeX
Concepts: Concurrency, REST APIs, Data Pipelines, Machine Learning, Bioinformatics